Jann Marco P. Cadiche BSIT 2-N OOP 03/03/2026

PART A - Variable Identification Table

Variable Name	Data Type	Variable Type	Stored IN
s1	Student	Reference Variable	Stack
s2	Student	Reference Variable	Stack
s3	Student	Reference Variable	Stack
name	String	Instance Variable	Heap
grade	int	Instance Variable	Heap
n	String	Parameter	Stack
g	int	Parameter	Stack
args	String[]	Parameter	Stack

Part B: Explanation. Please see example below

Statement	What Happens?	Memory Impact
Student s1 = new Student("John", 85);	Constructor is called; a new object is allocated	Stacks: s1 → Heaps: Object A (John, 85).
Student s2 = new Student("Maria", 90);	a second unique object is allocated.	Stack: s2 → Heap: Object B (Maria, 90).
Student s3 = s2;	s3 is pointed to the same memory address as s2	Stack: Both s2 and s3 point to object B.
s1.grade = 95;	The grade value of the first object is updated	Object A on the Heap now has a grade of 95.
s3.name = "Ana";	The name value of the object s3 points to is updated	Object B name changes to "Ana".

Part C - OUTPUT

John 95

Ana 90

Ana 90